

Packet Tracer – NAT and PAT Basic Configuration

Objectives

Part 1: Build a Small Network with Inside and Outside Networks

Part 2: Configure Static IP Addressing

Part 3: Configure NAT Overload (PAT)

Part 4: Verify Address Translation

Part 5: Test Connectivity to the Simulated Internet

Background / Scenario

Organizations commonly use private IP addresses internally and a public IP address to access external networks such as the Internet.

Because private addresses cannot be routed across the public Internet, routers perform **Network Address Translation (NAT)**. In most environments, **Port Address Translation (PAT)** is used, allowing multiple internal devices to share a single public IP address.

In this lab, you will configure PAT on a router and verify that multiple internal hosts can communicate with an external network using one public address.

This lab focuses on:

- Configuring NAT Inside and NAT Outside interfaces
 - Configuring NAT Overload (PAT)
 - Understanding private and public IP addresses
 - Verifying address translation tables
 - Testing connectivity through a translated address
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Instructions

Part 1: Build the Network Topology

Step 1: Add the Devices

a. Open **Cisco Packet Tracer**.

b. Add the following devices:

- 2 Routers
- 2 Switches
- 2 PCs
- 1 Server

Rename them:

- Router0 → R1
 - Router1 → ISP
 - PC0 → PC-A
 - PC1 → PC-B
 - Server0 → Web-Server
-

Step 2: Connect the Devices

Use **Copper Straight-Through** cables.

Connect:

PC-A → Switch1

PC-B → Switch1

Switch1 → R1 G0/0

R1 G0/1 → ISP G0/0

ISP G0/1 → Switch2

Switch2 → Web-Server

Your topology should resemble:

PC-A

\

Switch1
/
PC-B

|
|
R1
|
|
ISP
|
|
Switch2
|
Web-Server

Wait until all links turn green.

Part 2: Configure IP Addressing

Step 1: Configure PC-A

Navigate:

Desktop > IP Configuration

Enter:

IP Address: 192.168.10.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.10.1

Step 2: Configure PC-B

Enter:

IP Address: 192.168.10.20

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.10.1

Step 3: Configure Web Server

Enter:

IP Address: 209.165.200.10

Subnet Mask: 255.255.255.0

Default Gateway: 209.165.200.1

Step 4: Configure R1

Enter:

enable

configure terminal

interface g0/0

ip address 192.168.10.1 255.255.255.0

no shutdown

interface g0/1

ip address 209.165.201.2 255.255.255.252

no shutdown

end

Step 5: Configure ISP Router

enable

configure terminal

interface g0/0

ip address 209.165.201.1 255.255.255.252

no shutdown

interface g0/1

ip address 209.165.200.1 255.255.255.0

no shutdown

end

Part 3: Configure Routing

Step 1: Configure Default Route on R1

configure terminal

ip route 0.0.0.0 0.0.0.0 209.165.201.1

end

Step 2: Configure Route on ISP

configure terminal

ip route 192.168.10.0 255.255.255.0 209.165.201.2

end

Part 4: Configure PAT (NAT Overload)

Step 1: Identify Inside Network

Create an ACL.

configure terminal

access-list 1 permit 192.168.10.0 0.0.0.255

end

Step 2: Configure NAT Inside and Outside

configure terminal

interface g0/0

```
ip nat inside

interface g0/1
ip nat outside

end
```

Step 3: Configure PAT

```
configure terminal

ip nat inside source list 1 interface g0/1 overload

end
```

Explanation:

Inside Network:
192.168.10.0/24

Public Interface:
209.165.201.2

Overload:
Many devices share one public IP
This is PAT.

Part 5: Verify Connectivity

Step 1: Ping the ISP Router

From PC-A:

```
ping 209.165.201.1
```

Expected Result:

```
Reply from 209.165.201.1
```

Step 2: Ping the Web Server

From PC-A:

```
ping 209.165.200.10
```

Expected Result:

Reply from 209.165.200.10

Step 3: Test from PC-B

```
ping 209.165.200.10
```

Expected Result:

Reply from 209.165.200.10

Both PCs should successfully reach the external server.

Part 6: Verify NAT Operation

Step 1: View NAT Translations

On R1:

```
show ip nat translations
```

Example Output:

Pro	Inside Global	Inside Local
	209.165.201.2	192.168.10.10
	209.165.201.2	192.168.10.20

Observe:

- Inside Local = Private IP
 - Inside Global = Public IP
-

Step 2: View NAT Statistics

```
show ip nat statistics
```

Expected Information:

- Active translations
- Inside interfaces
- Outside interfaces
- Translation hits

Why PAT Is Important

Without NAT:

192.168.10.10

192.168.10.20

cannot travel across the Internet because they are private addresses.

With PAT:

192.168.10.10

192.168.10.20

↓

209.165.201.2

Many internal devices share a single public IP address.

This is how most home and enterprise networks access the Internet.

Troubleshooting Tips

If ping fails:

✓ Verify interfaces are up

show ip interface brief

✓ Verify routing table

show ip route

✓ Verify NAT configuration

show run | include nat

✓ Verify ACL configuration

show access-lists

✓ Verify translation table

show ip nat translations

✓ Confirm PCs have correct default gateways

✓ Confirm PAT uses the correct outside interface

What You Learned

- How NAT translates private addresses
- How PAT allows multiple hosts to share one public IP
- How to configure NAT Inside and NAT Outside interfaces
- How to verify NAT translations
- How routing and NAT work together
- How enterprises and home networks access the Internet

Challenge Extensions (Optional)

Challenge 1: Add a Third PC

Add:

PC-C

192.168.10.30

Verify:

ping 209.165.200.10

Check:

show ip nat translations

Challenge 2: Observe PAT in Simulation Mode

Switch to:

Simulation Mode

Generate pings from PC-A and PC-B.

Observe:

Private Address

↓

PAT Translation

↓

Public Address

↓

Destination Server

Challenge 3: Configure Static NAT

Create a permanent mapping.

Example:

```
ip nat inside source static 192.168.10.10 209.165.201.100
```

Verify:

```
show ip nat translations
```

Challenge 4: Add Another Internal Network

Create:

192.168.20.0/24

Add:

- New Switch
- New PC

Update ACL:

```
access-list 1 permit 192.168.20.0 0.0.0.255
```

Verify both networks can access the server through PAT.

Challenge 5: Break and Fix NAT

Remove NAT:

```
no ip nat inside source list 1 interface g0/1 overload
```

Test connectivity.

Questions:

- Can hosts still reach the server?
- Why not?
- Which NAT command restores functionality?